



CHRIST
UNIVERSITY
BENGALURU, INDIA

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CHRIST(Deemed to be University)

Department of Computer Science and Engineering

CSHO634CSP - INTRUSION DETECTION AND INCIDENT RESPONSE

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Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment.

Vision

Excellence and Service

Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

Unit 2 : Intrusion Detection System - Architecture & Tools

Centralized, Distributed, Cooperative Intrusion Detection, Tiered architecture, Intrusion detection in security, Tool Selection and Acquisition Process, Bro Intrusion Detection, Prelude Intrusion Detection, Cisco Security IDS, Snorts Intrusion Detection, NFR security, Architecture models of IDs and IPs.

Text Book :

Ali A. Ghorbani, Wei Lu, "Network Intrusion Detection and Prevention: Concepts and Techniques", Springer, 2010
(chapter 5)

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Topic 1 : Centralized Intrusion Detection Architecture

The **first generation of Intrusion Detection Systems (IDSs)** were **host-based** and operated *directly inside the mainframe or computer systems* they were designed to protect.

- These IDSs monitored the local system's activities such as operating system events, file access, user behavior, and processes.
- However, running an IDS directly on the target system added **significant overhead**—slowing down the system and consuming resources.

To address this limitation, later generations moved intrusion detection processes **off the target machine** onto a **separate system**, reducing load on the protected host.

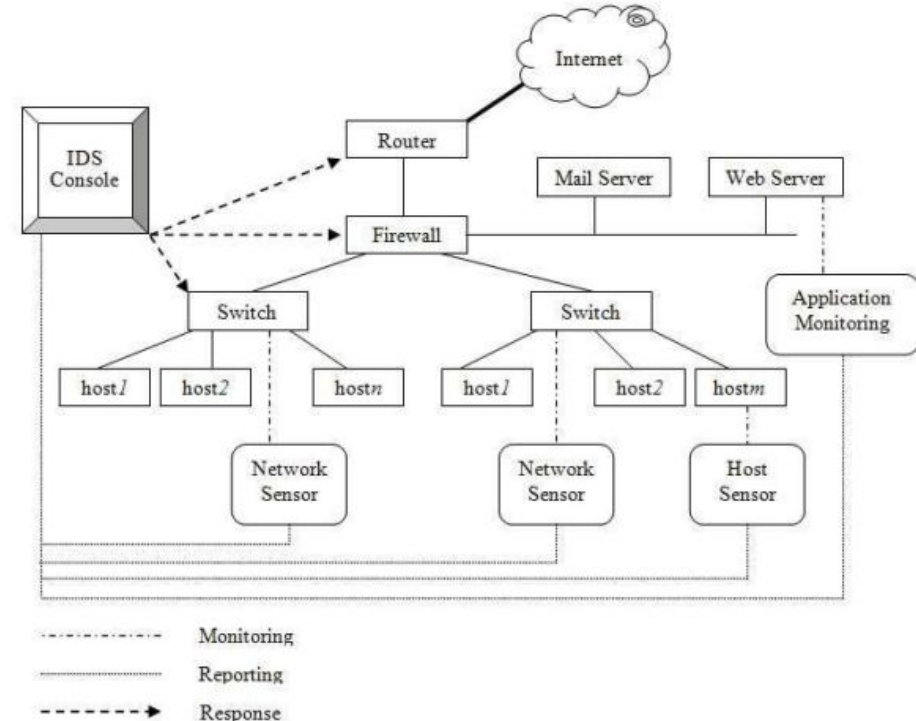


Fig. 5.1 Centralized IDS Architecture

contd...

Most modern IDSs today use a **centralized architecture**, where:

- All sensors or agents collect data from various hosts or network segments.
- A **central console** receives the information.
- All detection, alerting, logging, correlation, and response actions are managed **from this centralized server**.

Figure 5.1 (as referenced) typically shows:

Multiple hosts or network devices sending logs/events to **A central IDS server**, which performs **Analysis, signature matching, anomaly detection**, and **Generates alerts and responses**.

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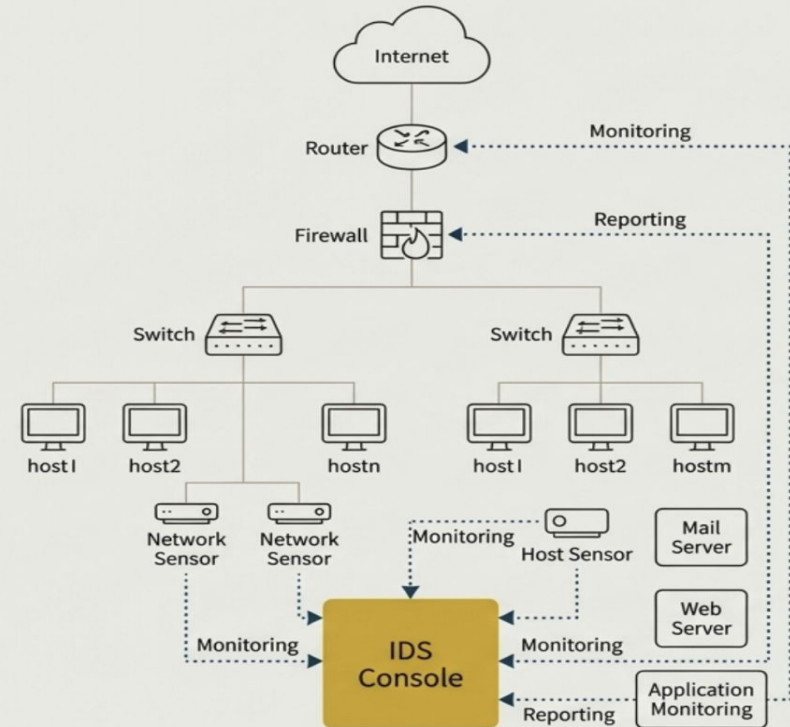
Topic 2: Distributed Intrusion Detection Architecture

The Limits of Centralized Defense

Traditional approaches to intrusion detection are centralized and face two major limitations:

1. **Incomplete Coverage:** Existing commercial solutions cannot cover all possible attacks on the network accurately (i.e., they drop packets, but generate a huge number of false alarms).
2. **Delayed Response:** Existing approaches are unable to respond to attacks in a timely manner.

Key Takeaway: A centralized system creates bottlenecks for analysis and single points of failure, struggling with the scale and speed of modern threats.



A Paradigm Shift: The Rise of Intelligent Agents

Instead of applying an individual IDS to defend the network, agent-based systems offer a new approach where several independent and intelligent processes cooperate to secure the network.



Distributed Computation

Reduces the processing burden on any single component.



Reduced Network Load

Decreases the amount of information sent over the network.



Platform Independence

Agents can operate across diverse systems.



Asynchronous Operation

Agents can act independently without constant coordination.



Extensibility & Scalability

New agents can be added easily to expand coverage.



Fault Tolerance & Resilience

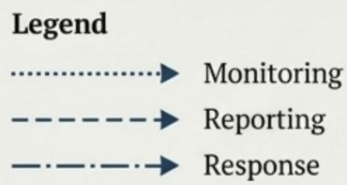
The system can withstand the failure of individual components.

This paradigm enables **Active Defense**: a modular, manageable way to implement an active defense instead of passive intrusion reporting.

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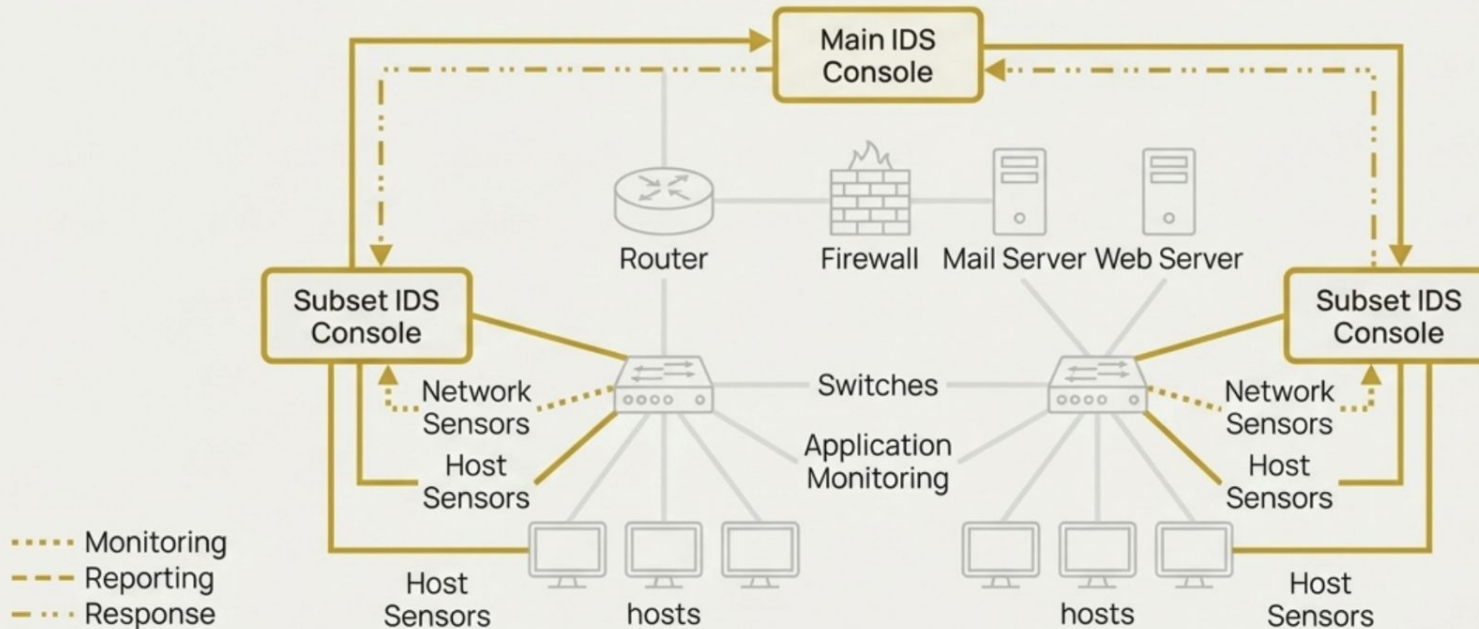
Legend

- Monitoring (dotted line with arrow)
- Reporting (dashed line with arrow)
- Response (dash-dot line with arrow)



Architectural Evolution II: The Hierarchical Model

The hierarchical architecture introduces an intermediate layer of "Subset IDS Consoles." These consoles manage local domains, analyzing data and handling responses for their specific area. They report up to a "Main IDS Console," reducing the load on the central system.





Agent Architectures in Practice

A closer look at six distinct approaches to building multi-agent intrusion detection systems.

The following systems showcase the diverse ways agent technology has been applied, each with unique agent roles, communication patterns, and detection strategies.

AAFID

MANSMA

Hummingbird

Hegazy et al.

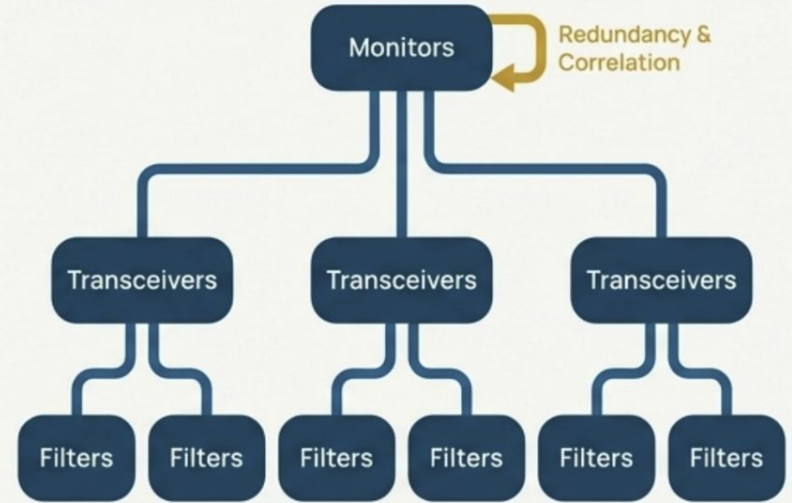
AHA! IDS

FAST

Autonomous Agents for Intrusion Detection (AAFID)

A distributed IDS developed at Purdue University where agents are organized hierarchically for data collection and analysis, enabling redundancy and fault tolerance.

Key Components / Agent Roles

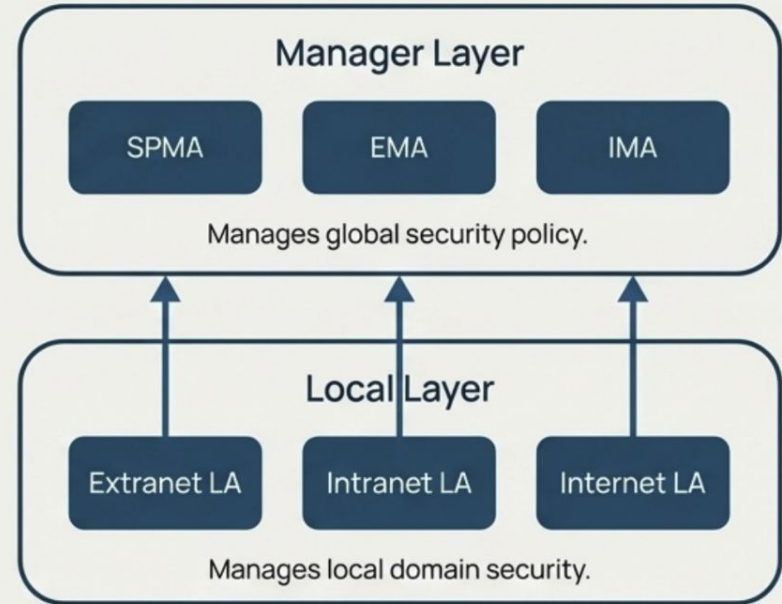


Distinguishing Feature: Its hierarchical monitor structure allows for both powerful, network-wide correlation and built-in redundancy, ensuring the system remains operational even if a monitor fails.

Multi-Agents System-based Network Security Management Architecture (MANSMA)

An architecture that applies Belief-Desire-Intention (BDI) agents across two distinct layers to manage global security policy and local domain security.

Key Components / Agent Roles

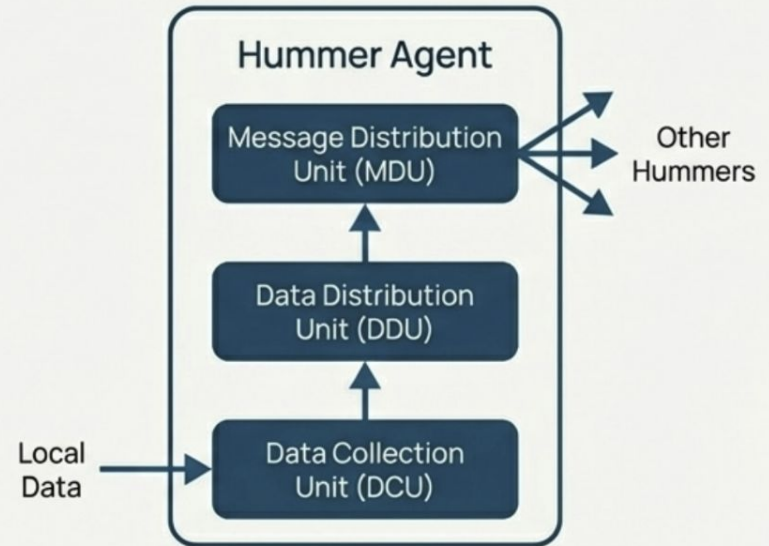


The explicit use of BDI agents, which model agent reasoning based on beliefs, desires, and intentions, allowing for more complex and rational decision-making.

Hummingbird

A distributed IDS that allows a system administrator to monitor security threats on multiple computers from a central console, while enabling data sharing across sites without compromising confidentiality.

Key Components / Agent Roles



Distinguishing Feature: Its biggest advantage is the ability to share data with other sites, enabling collaborative defense while minimizing the distribution of individual user hummer agents to protect privacy and system modularity.

Hegazy et al.'s Multi-agent-based IDS

A multi-agent IDS that classifies agents into four distinct categories based on their function, from simple data collection to complex decision-making.

Key Components / Agent Roles



Simple Reflex Agents:

Connect with networks to collect packets.

Analysis Agents:

Request log buffers to list suspicious packets.

Goal-based Agents:

Perform complementary analysis to make an intrusion decision.

Utility-based Agents:

Map the perceived state into a set of numbers that measure how closely goals are achieved.

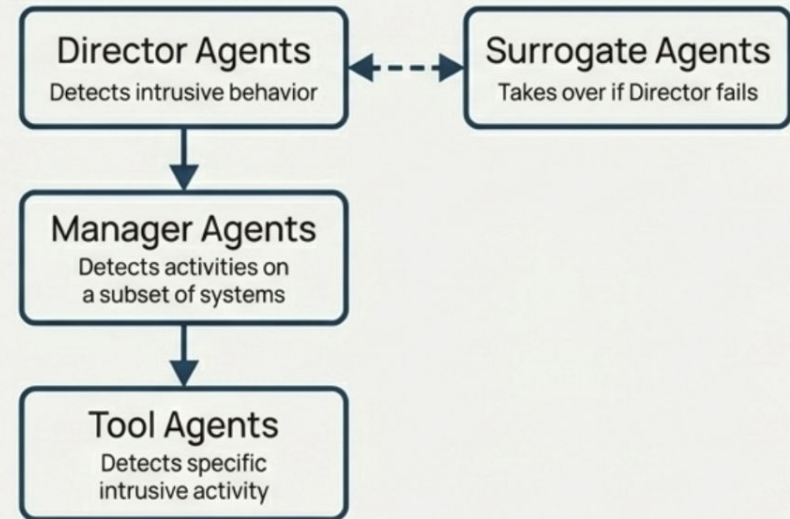
The clear taxonomy of agent roles, demonstrating a structured approach to distributing the complex task of intrusion detection among specialized software agents.

Adaptive Hierarchical Agent-based Intrusion Detection System (AHA! IDS)

On Source Serif Pro

A fully distributed, multi-agent framework that focuses on adapting its detection and response activities based on the perceived level of threat.

Key Components / Agent Roles



Distinguishing Feature: Its explicit focus on “types of detection adaption,” such as adjusting system resources, invoking new agent combinations, and adjusting low-level metrics in response to changing circumstances.

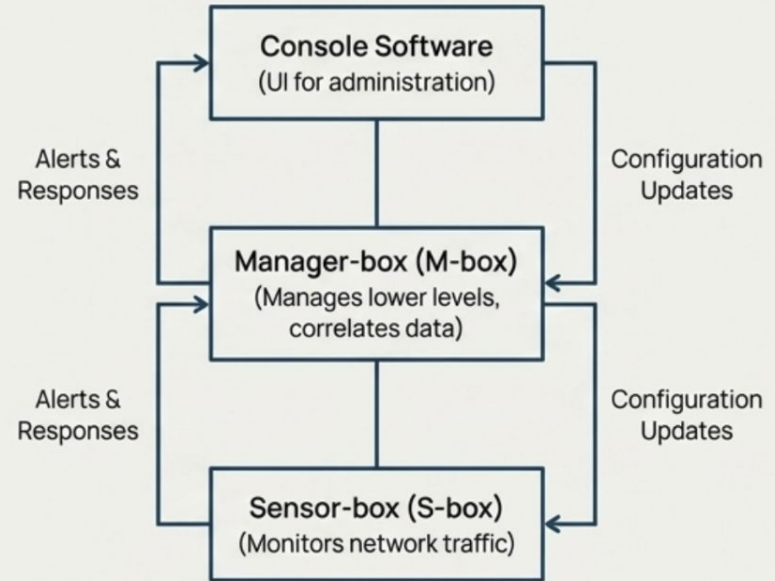
Fuzzy Adaptive Survivability Tools (FAST)

A multi-agent IDS that uses fuzzy logic to identify normal and abnormal patterns automatically, allowing it to handle the uncertainties associated with various attacks and network events.

Key Agent Roles:

- HCI Agents
- Monitor Agents
- Detection Agents
- Decision Agents

System Architecture



The application of fuzzy logic, which is used to identify the degree of suspicion for each attack and to deal with the uncertainties of response selection.

Common Threads in Agent-Based Defense

Across these diverse systems, several powerful, recurring principles emerge:

Role Specialization

Systems consistently divide the complex problem of intrusion detection among agents with highly specialized functions (e.g., data collection, analysis, decision-making, response).

- *Examples: Hegazy et al., AHA! IDS*

Collaborative Models

Whether through formal hierarchies or peer-to-peer networks, agents cooperate to build a comprehensive view of network state that no single component could achieve.

- *Examples: AAFID, MANSMA, Hummingbird*

Adaptive Response

Many systems move beyond simple alerting to dynamic, adaptive responses, reconfiguring themselves or adjusting resources based on the perceived threat.

- *Examples: AHA! IDS, FAST*

Resilience Through Redundancy

The distributed nature ensures that the failure of one or more agents does not lead to the failure of the entire security system.

- *Examples: AAFID, AHA! IDS*

Mobile Agent

The Inherent Challenges of a Mobile Agent Architecture

While promising, the practical implementation of mobile agents for security faces significant hurdles. Vigna's research highlights several key difficulties that any effective system must overcome.



Expensive

Difficult and costly to design and implement.



Complex

Difficult to develop, test, and debug.



Insecure Control

Difficult to authenticate and control agents securely.



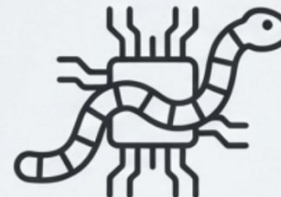
Hostile Environments

Agents are vulnerable to the malicious environments they execute in.



Lack of Standards

No ubiquitous infrastructure or shared language/ontology exists.



Exploitable

Can be easily exploited by worm attacks.

Two Pioneers, Two Philosophies

In response to these challenges, researchers have developed different architectural approaches. We will explore two seminal systems: IDA, which acts as a reactive digital detective, and MAIDS, which functions as a proactive intelligence refinery.

IDA: Intrusion Detection Agent System



"A reactive system that traces and verifies specific, suspicious events."

MAIDS: Mobile Agents for Intrusion Detection



"A hierarchical system that mines and refines network-wide data for a global view."

Deep Dive: IDA, The Digital Detective

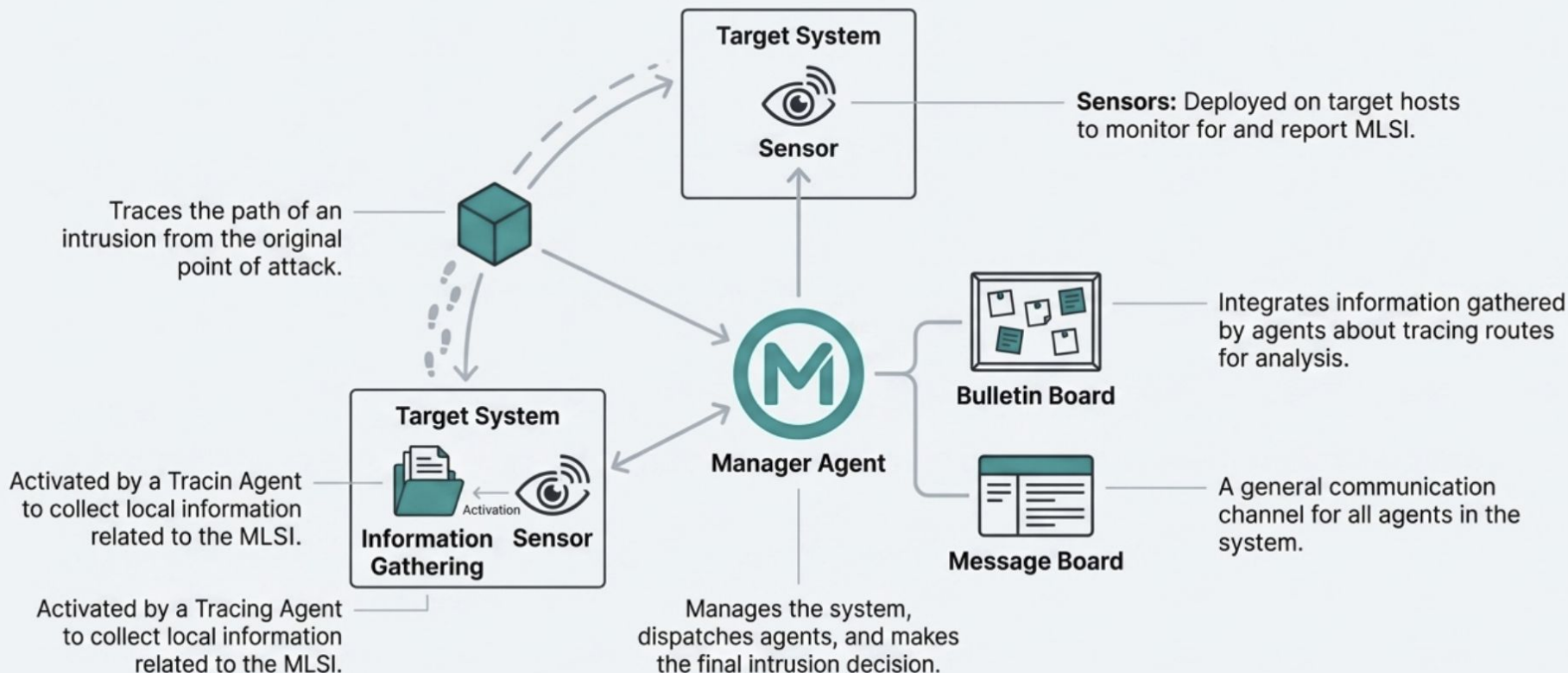
Core Philosophy

IDA's design focuses on *efficiently* detecting intrusions, rather than detecting *every possible* intrusion. It operates by investigating events that are only related to intrusions, triggered by specific "Marks Left by Suspected Intruder" (MLSI).

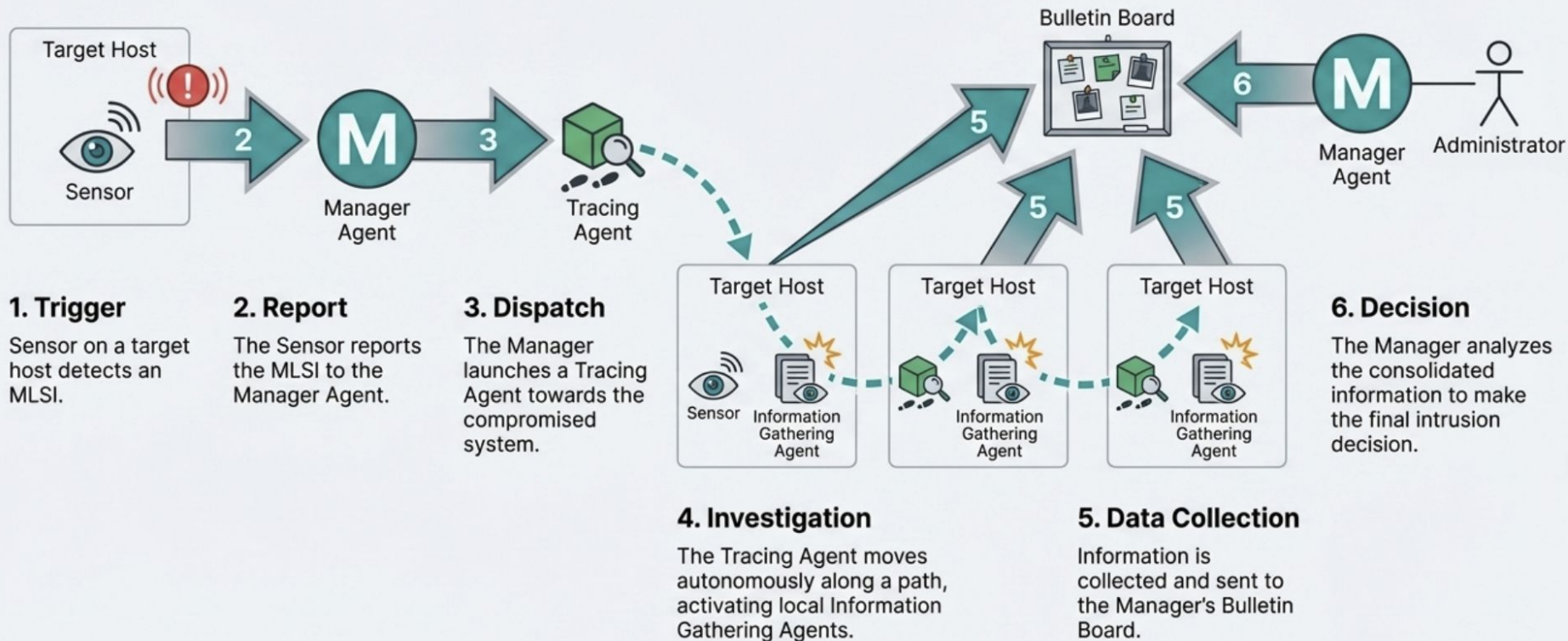
Key Features

- **Reactive:** Action begins when a sensor detects a potential MLSI.
- **Investigative:** Agents are dispatched to collect information along a suspected intrusion route.
- **Targeted:** Analysis is focused on specific hosts and events, not the entire network's data stream.

The Architectural Blueprint of IDA



IDA in Action: Anatomy of a Trace



Deep Dive: **MAIDS**, The Intelligence Refinery

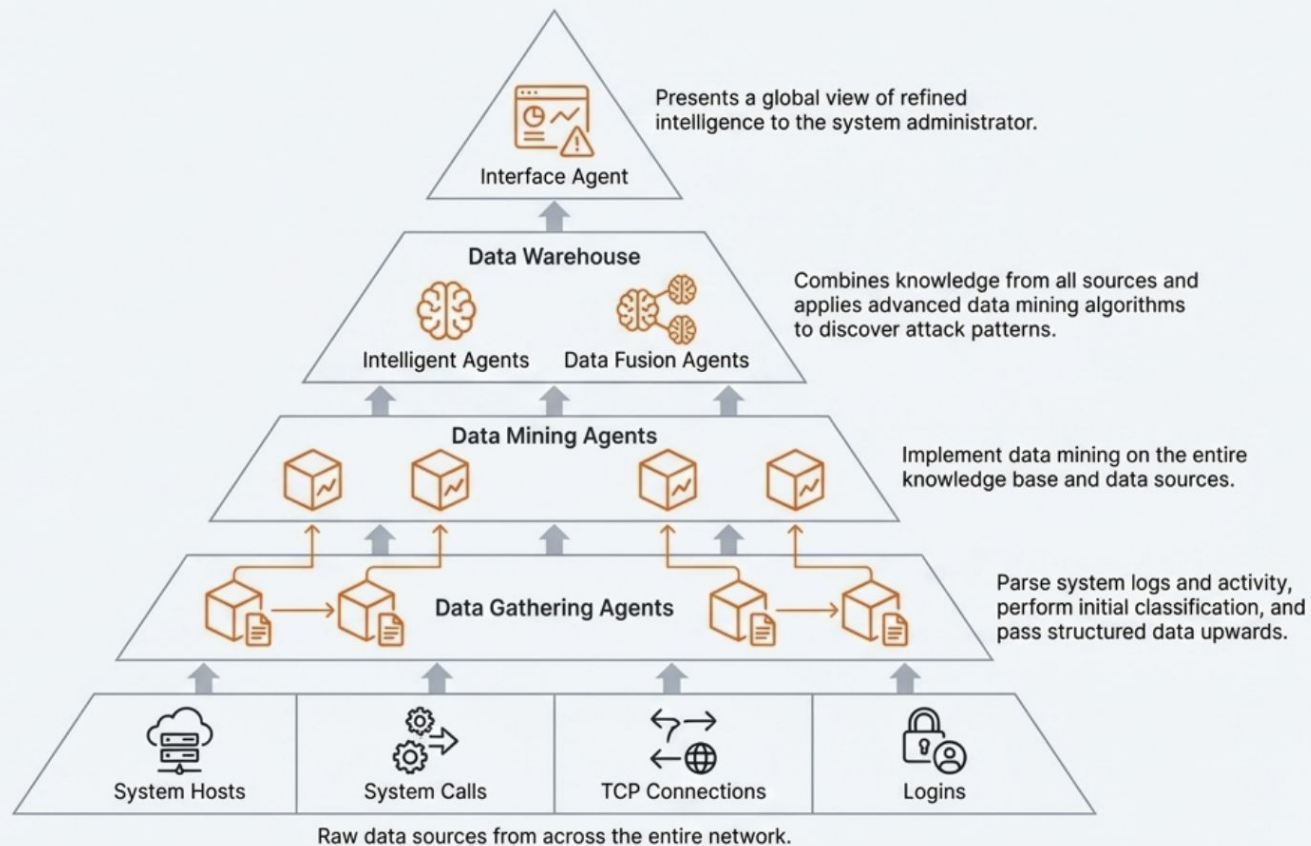
Core Philosophy

MAIDS is designed to provide a global and temporal view of the entire network. It uses distributed intelligent mobile agents and data mining techniques to discover associations among suspicious events and identify attack patterns.

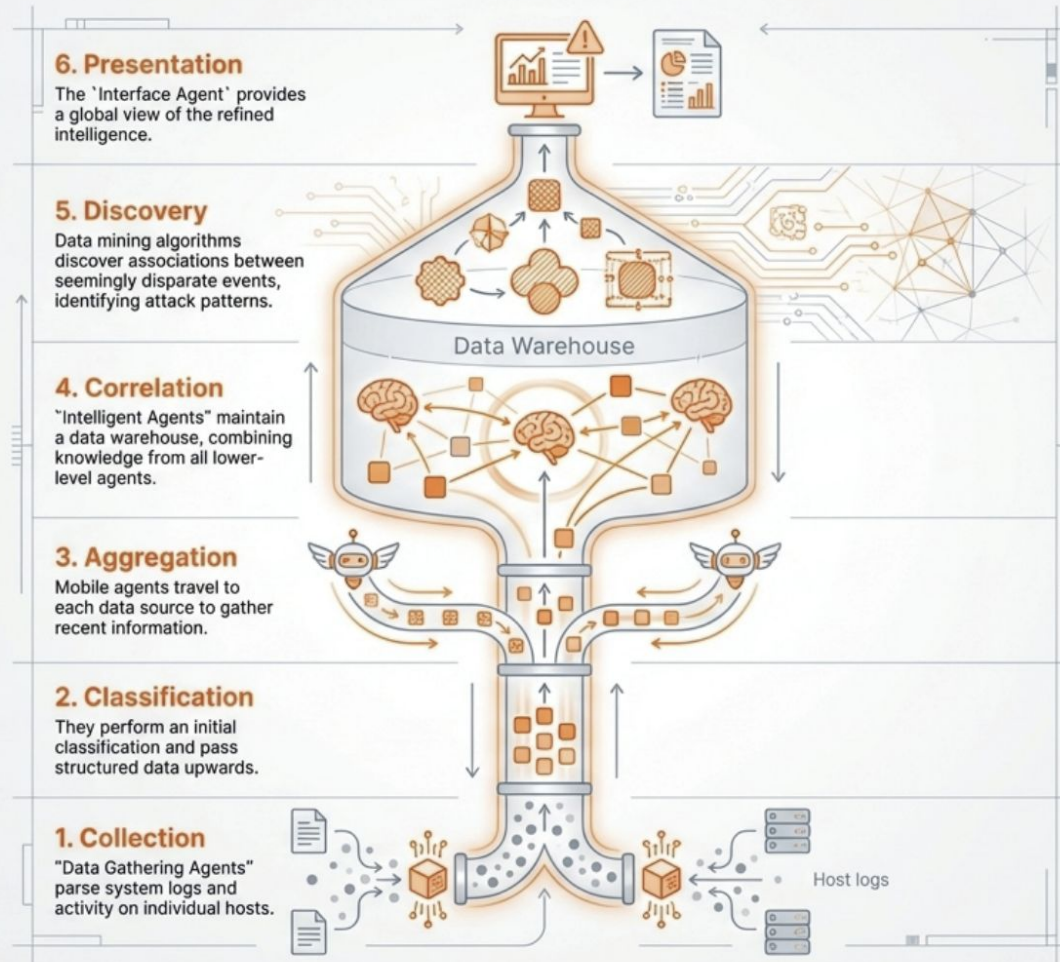
Key Features

- **Hierarchical:** Agents operate in layers, passing refined data upwards.
- **Lightweight:** Agents have minimal code and can be upgraded dynamically.
- **Data-centric:** Focuses on processing logs, system calls, and TCP connections to build a comprehensive knowledge base.

The Architectural Blueprint of **MAIDS**



MAIDS in Action: The Data Journey



Head-to-Head: A Tale of Two Architectures

Dimension	 IDA: The Digital Detective	 MAIDS: The Intelligence Refinery
Core Philosophy	Reactive Investigation: Traces and confirms specific threats (MLSI) after detection.	Proactive Aggregation: Mines network-wide data to build a global, temporal view.
Agent Structure	Team of Specialists: Distinct roles for managers, sensors, tracers, and gatherers.	Hierarchical Layers: Agents process and refine data as it moves up the structure.
Data Approach	Targeted & Deep: Collects specific information along a suspected intrusion route.	Broad & Correlated: Focuses on system calls, TCP connections, and logins network-wide.
Key Innovation	Autonomous Tracing: Eliminates the need to transfer large system logs to a server.	Data Mining & Lightweight Agents: Enables discovery of complex patterns with dynamic agents.

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Cooperative Detection Architecture

The Blueprint for Effective Cooperation

Effective cooperation is not simply about connecting systems. It requires a foundational framework built on principles of trust, autonomy, resilience, and intelligent data handling.

The following ten basic principles form the blueprint for designing and implementing a robust cooperative intrusion detection architecture.

The Principles of Governance & Trust



Local Control Over Policy: Each cooperating network retains control over its own policy decisions.



Local Control Over Sharing: The local network decides which information to provide to other parties.



Data Authenticity and Integrity: The authenticity and integrity of information must be provable, as a source can be compromised.



Local Policy Enforcement: Hosts and networks that identify policy violations are not responsible for enforcement elsewhere; enforcement remains local.

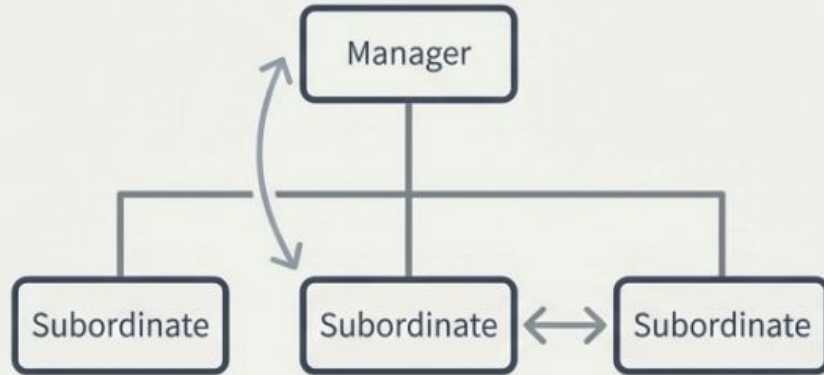


Mandatory Authentication: A necessary authentication mechanism must exist between all cooperating networks.

The Principles of Architecture & Resilience

Principle 6

Hierarchical Structure: A hierarchical architecture of cooperative entities is necessary. Information sharing can be vertical (manager to subordinate) or horizontal (between subordinates).



Principle 7

Redundant Data Collection: Data collection should be redundant so the system can continue operating even when one data collector is compromised or becomes unavailable.

The Principles of Data & Intelligence



Principle 8: Data Reduction

Data reduction is necessary to avoid exchanging high volumes of data among partners.



Principle 9: Data Sanitization

Sanitization is required to mitigate security risks caused by sharing critical host and network-specific attributes.



Principle 10: Data Visualization

In a large-scale framework, visualization tools are required for human system/network administrators to analyze the data in a timely manner.

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Tiered Architecture

- Tiered Intrusion Detection System (IDS) Architecture organizes intrusion detection components into multiple layers (tiers), where each tier performs a specific role in detecting, analyzing, and responding to security threats.
- Instead of having one IDS do everything, the workload is divided across levels.
- At least three types of tiered architectures can be used:
 - Single-tiered architecture
 - Multi-tiered architecture
 - Peer-to-peer architecture

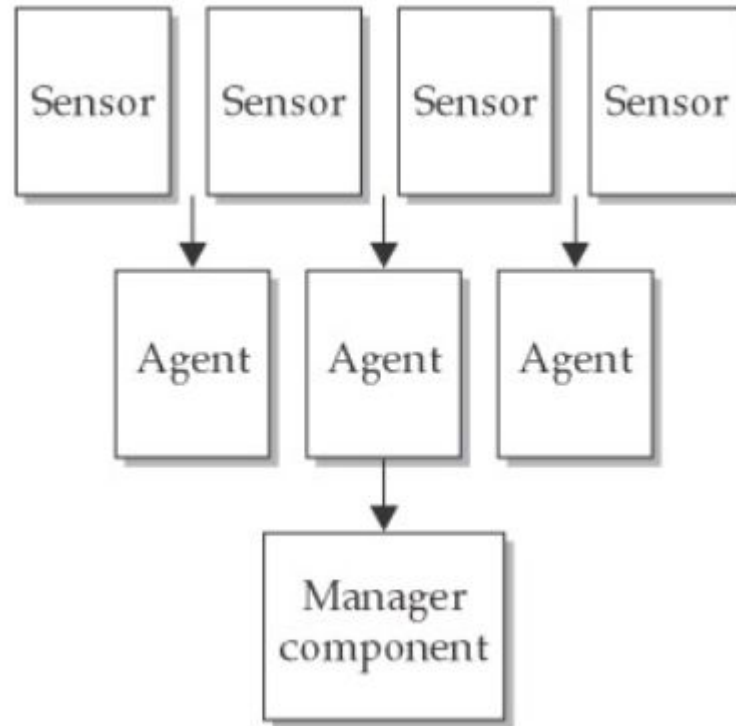
1. Single Tiered Architecture

- Components in an IDS collect and process data themselves, rather than passing the output they collect to another set of components.
- **An example of a single-tiered architecture** is a host-based intrusion-detection tool that takes the output of system logs and compares it to known patterns of attack.
- A single tier offers advantages, such as simplicity, low cost, and independence from other components.
- At the same time, however, a single-tiered architecture usually has **components that are not aware of each other**, reducing considerably the potential for efficiency and sophisticated functionality.

2. Multi Tiered Architecture

- A multi-tiered architecture involves multiple components that pass information to each other.
- Many of today's IDSs, for example, consist of three primary components:
 - Sensors
 - Analyzers or agents
 - Manager
- Sensors perform **data collection**.
- For example, network sensors are often programs that capture data from network interfaces.

Multi Tiered Architecture



Multi Tiered Architecture

- Sensors can also collect data from system logs and other sources, such as personal firewalls and TCP wrappers.
- Sensors pass information to agents (sometimes also known as analyzers), which monitor intrusive activity on their individual hosts.
- Each sensor and agent is configured to run on the particular operating environment in which it is placed.
- Agents are normally specialized to perform one and only one function.
- One agent might, for example, examine nothing but TCP traffic, whereas another might examine only FTP (File Transfer Protocol) connections and connection attempts.

Multi Tiered Architecture

- When an agent has determined that an attack has occurred or is occurring, it sends information to the manager component, which can perform a variety of functions including (but not limited to) the following:
 - Collecting and displaying alerts on a console
 - Triggering a pager or calling a cellular phone number
 - Storing information regarding an incident in a database
 - Retrieving additional information relevant to the incident
 - Sending information to a host that stops it from executing certain instructions in memory

Multi Tiered Architecture

- Sending commands to a firewall or router that change access control lists
- Providing a management console—a user interface to the manager component
- A central collection point allows for greater ease in analyzing logs because all the log information is available at one location.
- Additionally, writing log data to a different system from the one that produced them is advisable; if an attacker tampers with or destroys log data on the original system, the data will still be available on the central server—the manager component.

Multi Tiered Architecture - Advantages

- Advantages of a multi-tiered architecture include greater efficiency and depth of analysis.
- It can also provide a much more complete picture of the security condition of an organization's entire network and the hosts therein, compared to a single-tiered architecture.
- The main downsides include increased cost and complexity.
- The multiple components, interfaces, and communications methods translate to greater difficulty in setting up this architecture and more day-to-day maintenance and troubleshooting challenges.

Peer to Peer Architecture

- It involves exchanging intrusion-detection and intrusion-prevention information between peer components, each of which performs the same kinds of functions.
- **This peer-to-peer architecture is often used by cooperating firewalls.**
- As one firewall obtains information about events that are occurring, it passes this information to another, which may cause a change in an access control list or addition of restrictions on proxied connections.
- The second firewall can also send information that causes changes in the first.
- Neither firewall acts as the central server or master repository of information.

Peer to Peer Architecture

- The main advantage of a peer-to-peer architecture is simplicity.
- Any peer can participate in what is effectively a group of peer machines, each of which can benefit from the information the others glean.
- The main downside is a lack of sophisticated functionality due to the absence of specialized components.

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Intrusion Detection in Security

- Intrusion detection is the process of monitoring systems and networks to identify:
 - Unauthorized access
 - Misuse of resources
 - Policy violations
- It focuses on detection, not prevention
- IDS watches and alerts when something suspicious happens.

Intrusion Detection in Security

Importance:

- Firewalls and authentication cannot stop all attacks
- Attacks can originate from:
 - Insiders
 - Compromised accounts
 - Zero-day exploits
- IDS acts as a second line of defense

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Tool Selection and Acquisition Process

- An Intrusion Detection System (IDS) is a critical security component used to monitor networks and hosts for malicious activities.
- Selecting the right IDS tool is essential because an inappropriate tool can lead to **excessive false alarms, poor detection capability, or high operational costs.**
- Therefore, IDS tool selection must follow a systematic and well-defined acquisition process.

Tool Selection and Acquisition Process

- Different organizations face different security threats.
- IDS tools vary in detection techniques, architecture, scalability, and cost.
- Incorrect selection may result in:
 - Missed attacks
 - Alert fatigue
 - Inefficient incident response
- Proper selection ensures effective security monitoring and optimal resource utilization.

Tool Selection and Acquisition Process

Defining Security Requirements:

- Before selecting an IDS tool, organizations must clearly define their security needs:
 - Network size and structure (small, enterprise, cloud-based)
 - Critical assets (databases, servers, applications)
 - Threat landscape (external attacks, insider threats, malware)
 - Regulatory and compliance requirements
 - Response expectations (alert-only or automated actions)

Tool Selection and Acquisition Process

Selection of IDS Type:

- Based on requirements, the appropriate IDS type must be identified:
 - Host-Based IDS (HIDS) – monitors individual systems
 - Network-Based IDS (NIDS) – monitors network traffic
 - Hybrid IDS – combines host and network monitoring
 - Centralized, Distributed, or Tiered architecture
 - Real-time or Offline detection
- This step ensures alignment between organizational needs and IDS capability.

Tool Selection and Acquisition Process

Performance and Scalability

- Performance evaluation ensures the IDS can handle real-world conditions:
 - Maximum traffic throughput
 - Impact on system performance
 - Ability to scale with network growth
 - Handling of peak traffic without packet loss

Tool Selection and Acquisition Process

Compatibility and Integration

- An IDS must integrate seamlessly with existing security infrastructure:
 - Firewalls and IPS
 - SIEM platforms
 - Log management systems
 - Operating systems and network devices
- Good integration reduces administrative overhead and improves **visibility**.

Tool Selection and Acquisition Process

Cost Considerations

- Tool selection must consider total cost of ownership, not just license cost:
 - Software licensing or subscription fees
 - Hardware requirements
 - Training and skill development
 - Maintenance and operational costs
- Open-source tools may reduce licensing costs but require skilled administrators.

Tool Selection and Acquisition Process

Tool Acquisition:

- After evaluation, finalize the IDS tool
- Acquire through:
 - Commercial vendors
 - Open-source platforms
- Document deployment scope, policies, and configurations

Tool Selection and Acquisition Process

Deployment and Training

- Effective deployment requires:
 - Proper sensor placement
 - Rule tuning and configuration
 - Training SOC staff
 - Defining escalation and incident response procedures
- Human expertise is critical for IDS success.
- **Continuous Monitoring and Review**

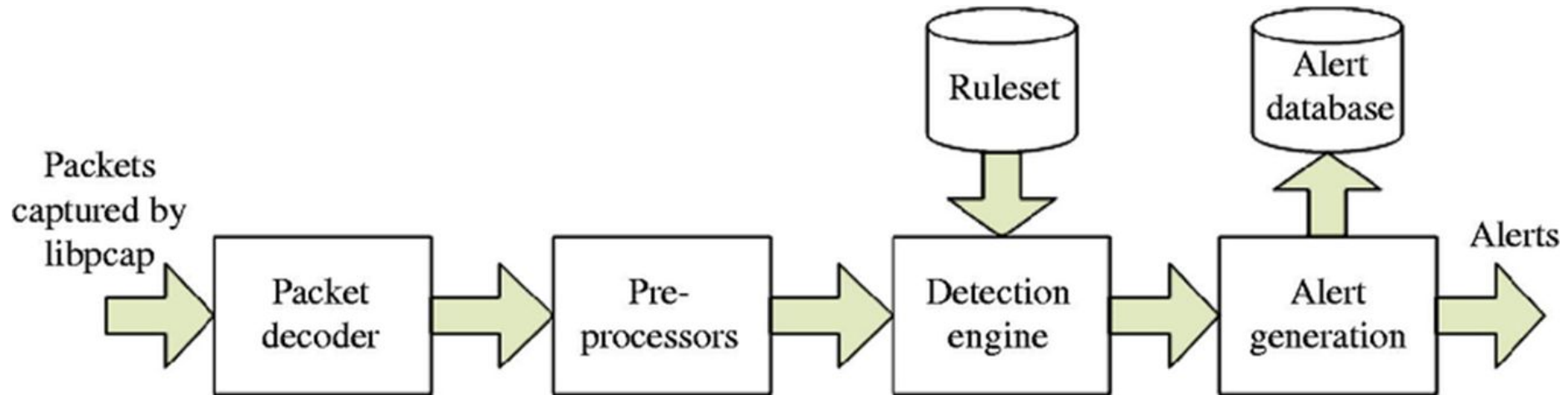
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Snort Intrusion Detection

- Snort is based on library packet capture (libpcap).
- Libpcap is a tool that is widely used in Transmission Control Protocol/Internet Protocol address traffic sniffers, content searching and analyzers for packet logging, real-time traffic analysis, protocol analysis and content matching.
- Users can configure Snort as a sniffer, packet logger – like TCPdump or Wireshark – or network intrusion prevention method.

Snort Architecture



Snort Intrusion Detection

Packet logger and sniffer mode:

- If a subscriber configures Snort to operate as a sniffer, it will scan network packets and identify them. Snort can also log those packets to a disk file.
- To use Snort as a packet sniffer, users set the host's network interface to promiscuous mode to monitor all network traffic on the local network interface.
- It then writes the monitored traffic to its console.
- By writing desired network traffic to a disk file, Snort logs packets.

Snort Intrusion Detection

Intrusion prevention system mode:

- As an open source network intrusion prevention system, Snort will monitor network traffic and compare it against a user-defined Snort rule set – the file would be labeled `snort.conf`. This is Snort's most important function.
- Snort applies rules to monitored traffic and issues alerts when it detects certain kinds of questionable activity on the network.

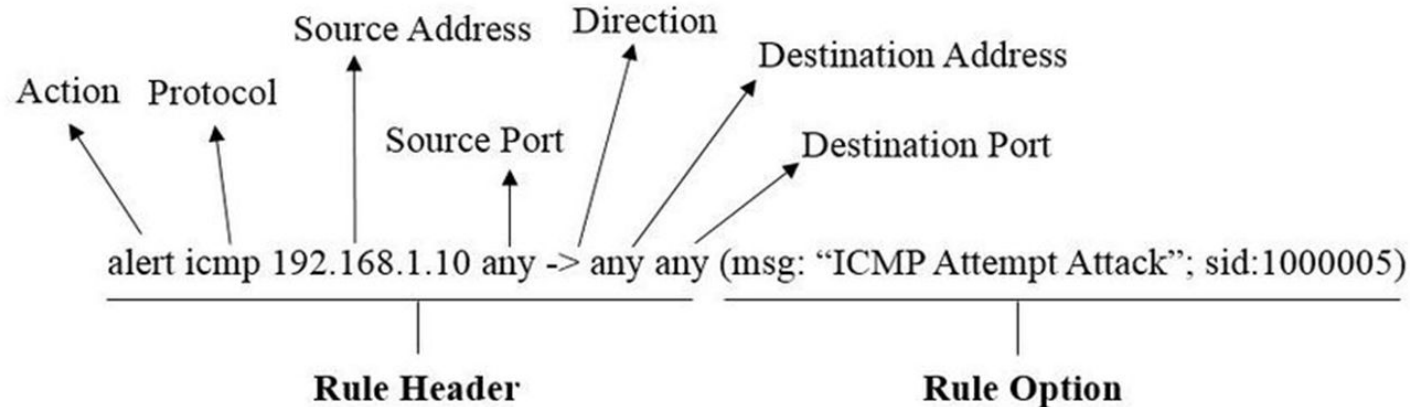
Snort Intrusion Detection

Rules and configuration:

- Alert rules: This generates alerts using the alert method.
- Log rules: Upon generation of any alert, it logs that specific alert.
- Pass rules: Ignores the packet if deemed malicious and drops it.

Snort Intrusion Detection

Rules Example:



Snort Intrusion Detection

Steps for Installing Snort:

- Install Npcap
- Install Snort for windows from [snort.org](https://www.snort.org).
- Install rules from [snort.org](https://www.snort.org)
- Community, Registered and Subscribed rules
- Initial configuration of `snort.conf` file
- Run snort in command prompt

Thank you